Shrey Singh

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Bengaluru, India

Summary

Software engineer with machine learning experience at NetApp and Agneyas Labs, building forecasting pipelines and internal ML tools. Builds complete products end to end, from React and Next.js front ends to Python and Node.js back ends, databases, and deployment.

Experience

- **Agneyas Labs** – *Bengaluru, India* *Jul 2026 – Present*
  - Built and deployed an internal AutoML platform (Angular, Flask) that trains and benchmarks multiple algorithms per dataset, surfaces the strongest performer, and lets users run predictions and persist both models and results; in use by an internal team of 10–12.
  - Developing a hybrid battery digital twin on proprietary EV cycling data: an equivalent-circuit model constrains known physics (terminal voltage, internal resistance, capacity fade) while a learned residual network absorbs nonlinear aging, cell-to-cell variance, and thermal effects.
- **NetApp** – *Bengaluru, India* *Jun 2025 – May 2026*
  - Built and deployed a one-click forecasting pipeline for $150M+ quarterly revenue planning using 1.5M+ records.
  - Improved sales commission forecast accuracy by 15% with an XGBoost ensemble, reducing MAPE to 6–10% for GTM financial accruals across 4,900+ sales representatives.
  - Designed scalable ML workflows for preprocessing, training, validation, and batch inference, reducing end-to-end runtime from 6–8 hours to 2 hours.

Projects

- **Argus: Real-Time Fraud Detection Engine** *Code*
  Tools: Python, XGBoost, Scikit-learn, SHAP, FastAPI, Next.js, Docker
  - Built a real-time fraud scoring engine combining supervised XGBoost with an unsupervised Isolation Forest into one 0–100 risk score, with SHAP explanations showing why each transaction was flagged.
  - Used the same feature code for training and live scoring, evaluated on a time-based split (train on the past, test on the future) to avoid data leakage, and streamed scores to a Next.js dashboard over WebSocket.
- **RL Statistical Arbitrage Engine** *Code*
  Tools: Python, Stable-Baselines3, Gymnasium, statsmodels, hmmlearn, MLflow
  - Built a pairs-trading backtester that picks stock pairs whose prices move together (cointegration tests) and trains RL agents (PPO, SAC) to decide when to enter, exit, and how much to trade, after transaction costs.
  - Evaluated with walk-forward testing (retrain on each past window, test on the next) against rule-based baselines, using market-regime features from a Hidden Markov Model, MLflow experiment tracking, and 94 tests.
- **RAG Knowledge Assistant: Chat with Your Documents** *Code*
  Tools: Python, FastAPI, PostgreSQL, pgvector, LangChain, OpenAI, Next.js, Docker
  - Built a document Q&A app: uploaded PDFs are split into chunks, embedded, and stored in PostgreSQL with pgvector, and an LLM answers questions only from the retrieved passages, with citations.
  - Combined vector and keyword search for better retrieval, logged latency and token cost for every query, and added an evaluation page that scores answer faithfulness and citation coverage; 65 tests.
- **Email Campaign Scheduler** *Code*
  Tools: TypeScript, Node.js, Express, BullMQ, Redis, PostgreSQL, Prisma, Next.js
  - Built a bulk email scheduler where users upload recipients and pick a start time; each email becomes a delayed job in a Redis-backed queue, sent by a separate worker, with a Next.js dashboard and Google sign-in.
  - Enforced per-sender hourly sending limits with Redis counters and used idempotency keys so emails are never lost or sent twice, even when the server restarts mid-campaign.

Education

- **Manipal Institute of Technology** *Manipal, India*
  B.Tech in Computer Science and Engineering, Specialization in AI/ML | CGPA: 8.22 2022 – 2026

Skills

- **Languages:** Python, TypeScript, JavaScript, SQL, Java, C++, C, Go, HTML, CSS
- **Web and Backend:** React, Next.js, Node.js, Express, FastAPI, Flask, Angular, REST APIs, WebSockets
- **ML and AI:** PyTorch, Scikit-learn, XGBoost, SHAP, Stable-Baselines3, LangChain, RAG, Pandas, NumPy
- **Data Science:** Time-Series Forecasting, Reinforcement Learning, Feature Engineering, Anomaly Detection
- **Databases and Tools:** PostgreSQL, SQLite, Redis, Docker, Git, Linux, MLflow, CI/CD, Vercel